TERRAPROOF | SATELLITE EVIDENCE DOSSIER

Autonomous Geospatial Audit & Decision-Support System • Generated 2026-10-28 07:27:25 UTC

Audit Dossier ID:	35cb5a2f-4eaa-4608-b759-3a9a72f63e13	**Analysis Type:**	FLOOD DETECTION
Claim Reference:	N/A	**Crop Context:**	N/A
Farmer / Insured:	N/A	**Recommended Verdict:**	**SIGNIFICANT_INUNDATION_OBSERVED**

1. EXECUTIVE EVIDENCE FINDINGS

Detected 2757.232 ha (100.0% of study area) showing active surface inundation following the precipitation event.

Metric	Value	Methodological Meaning
Baseline Mean Index	-0.782	Mean radiometric canopy baseline prior to reported event
Post-Event Mean Index	0.5477	Mean radiometric response observed immediately following event
Index Difference (Delta)	1.3297	Magnitude of spectral divergence (positive = loss/inundation)
Estimated Affected Area	**2757.232 ha**	Total hectares showing thresholded anomalous change
Claimed Loss Area	N/A ha	Hectares reported in submission form
Confidence Score	**85%**	High contrast between near-infrared absorption and green reflectance typical of standing water.

2. METEOROLOGICAL CONTEXT (OPEN-METEO ARCHIVE)

Ground coordinates (26.875, 75.775) recorded **68.4 mm** total rainfall during the target window, peaking at 41.2 mm/day. Average topsoil moisture was estimated at 0.38 m³/m³.

3. SATELLITE SENSORS & ACQUISITIONS USED

Scene ID	Sensor	Acquisition Date	Cloud Cover
DEMO_S2A_MSIL2A_20260701_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-01T10:30:00Z	4.2%
DEMO_S2B_MSIL2A_20260701_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-01T10:35:00Z	8.5%
DEMO_S2A_MSIL2A_20260716_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-16T10:30:00Z	4.2%
DEMO_S2B_MSIL2A_20260716_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-16T10:35:00Z	8.5%

4. POTENTIAL INCONSISTENCIES & AUDIT FLAGS

• No major numerical discrepancy flagged between claimed parameters and spectral observations.

5. METHODOLOGICAL LIMITATIONS & RESPONSIBLE USE

- Dense vegetation canopies may mask standing water underneath (emergent vegetation limit).
- Smooth surfaces such as tarmac or flat rocks can produce low SAR backscatter similar to water.
- Cloud shadows can mimic water in single-date optical scenes; pre-post differencing mitigates this.

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